

polysilicon production and reduction in material usage.

Signs of innovations and improvements in these areas are already visible. Today, silicon usage is down to 10 gm/Wp, which until a few years ago was typically 13 gm/Wp. There is significant potential for improvement of manufacturing processes in the near future. The European Union (EU) is targeting polysilicon consumption below 5, 3 and 2 gm/Wp in the short, medium and long term, respectively.

According to available market research, crystalline silicon cell may touch \$1.00/Wp in the EU by 2012.

Government incentives

The incentive structure is currently offered under the Special Incentive Package Programme of the Semiconductor Policy. The policy, which encourages semiconductor and solar PV manufacturing, offers a capital subsidy of 20 per cent for manufacturing plants in SEZs and 25 per cent for manufacturing outside SEZs. The subsidy, however, is based on the condition that the net present value (NPV) of

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integrated manufacturing base in the country.

2. The availability of capital subsidy would ensure early capital recovery or break-even for the investors and allow the investors to commit higher investments into this sector.

3. Emphasis should be laid on R&D and innovation in solar photovoltaics, as these are one of the key drivers for the development of the solar PV industry. The salient initiatives in this direction include collaborative research amongst government, R&D institutions and industry, enhancing coordination amongst various government departments and institutes undertaking R&D, commercialisation of the developed

ing' to all grid-connected consumers generating solar power will incentivise all consumers to adopt solar PV.

7. Provision of financial assistance at cheaper rates to both, the manufacturers and the developers, will also enhance the competitiveness of this sector and greatly help in achieving the grid parity through solar PV.

8. Besides large-scale applications, the government should encourage solar PV applications for small- and medium-scale niche market segments (residential, commercial and telecom).

Application focus

According to Dr Kaul, the production and R&D should be targeted at rural applications where people are deprived of light—the basic necessity of life. He feels that the best application of a PV module in rural areas is the water-pumping system. As the pumping system doesn't involve any electronics and has no battery in between, it can be repaired by any plumber in the village.

The target applications for solar cells developed by CEL are power plants for feeding into the grid and decentralised PV systems for rural village electrification. Under a Rural Village Electrification (RVE) programme in Bastar area of Dantewada district in Chattisgarh, where people don't know what artificial light is, CEL provided solar light in the form of household and streetlights and portable solar lanterns. 35.5cm B&W TVs installed at central places now run on solar power without inverter. ●

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“Rural electrification and telecom would be the major drivers of growth.—Ajay Prakash Shrivastava, president, Maharishi Solar Technology



the investment is at least \$212 million. This has resulted in investors showing interest to set up large-scale vertically integrated manufacturing facilities.

It is crucial to implement the incentive package fast so that India can establish a manufacturing base of a commendable scale. As would be seen in the future, the manufacturing base has to be adequately supported by the capital subsidy programme.

Expectations from the government

1. Significant and immediate steps are required from the government of India to facilitate a bigger and vertically

technology and developing a proper framework for technology transfer, and collaboration within India and other countries to obtain the best available technology.

4. On the deployment side, it is recommended that the government extends the GBI scheme to all project developers for unlimited capacity addition in the next five years.

5. To accelerate the demand, the government should enact a Renewable Energy Law requiring all utilities to progressively increase their purchase of power (year after year) from the RE segments.

6. Government nod for 'net meter-